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Wood-Based Hybrid Construction and Circular Economy: A Three-Level Framework for Component, System, and Building Design

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Abstract

Hybrid construction is a key strategy for increasing the use of wood in contemporary buildings while meeting structural and regulatory requirements, particularly as building-level carbon reporting and embodied-carbon requirements are becoming increasingly important in Finland and other regulatory contexts. This study aims to examine how wood-based hybrid construction can be systematically structured to support low-carbon and circular building practice, distinguishing single component level, system level, and building level strategies. This research builds on a research, development, and education learning materials project at Tampere University and adopts a qualitative, non-experimental methodology based on a structured synthesis of peer-reviewed literature, technical sources, and documented application examples. The main contribution of this study is a three-level decision-support framework that clarifies how wood-based hybridization can be evaluated across different design scales in relation to structural performance, circular economy mechanisms, and early-stage design decision-making. The results suggest that hybridization effectiveness is scale-dependent, that higher hybridization levels may provide broader strategic opportunities for regulatory alignment and lifecycle-oriented design when supported by appropriate system configuration, and that circular economy performance depends on connection logic, adaptability, and material separability rather than material choice alone. The findings provide a practical reference for architectural design, professional practice, and education.

1. Introduction

The construction sector remains one of the most resource-intensive and environmentally impactful industries worldwide [1-6]. According to consolidated international assessments, buildings and construction activities account for approximately 36-40% of global energy consumption, around 36–39% of energy-related CO_2 emissions, and up to 40% of total material-related waste generation [7,8]. While regulatory efforts and technological advances have significantly reduced operational energy demand in new buildings over the past two decades, the relative share of embodied carbon associated with construction materials has increased markedly [9]. Recent studies indicate that embodied emissions can represent 30-50% of total lifecycle emissions in contemporary low-energy and nearly zero-energy buildings, particularly in urban contexts [10].

Within this shifting environmental balance, structural systems and material choices have become critical leverage points for emission reduction [10-12]. This development has renewed interest in bio-based construction materials, particularly wood and engineered wood products (EWPs), as alternatives to emission-intensive mineral-based systems [13].

1.1. Wood in Hybrid Construction: Conceptual and Historical Context

Wood has been used as a primary construction material for millennia; however, its systematic combination with other structural materials emerged more clearly during the industrial period. Early forms of timber–steel hybridization, including flitch beams, emerged during the nineteenth century as a means of increasing bending capacity while preserving the practical advantages of timber construction [14]. The first systematically documented timber–concrete

composite floor systems appeared in the 1930s, notably with mechanically connected systems developed to improve stiffness and vibration performance in long-span floors [15].

A significant technological shift occurred from the 1990s onward with the industrialization of EWPs such as glued laminated timber (glulam) and laminated veneer lumber (LVL), followed by the wider adoption of cross-laminated timber (CLT) in the early 2000s [16]. These developments enabled timber to be used reliably in mid-rise and, later, high-rise buildings. However, pure timber solutions continued to face limitations related to fire resistance, serviceability, span capability, and regulatory acceptance, particularly beyond 6-8 stories in many national codes [17].

As a result, hybrid construction gained prominence as a pragmatic response rather than an ideological preference. Contemporary hybrid systems deliberately combine timber with concrete, steel, or composite materials at component, system, or building scale [18]. This approach allows each material to be positioned according to its mechanical and physical strengths, rather than forcing timber to perform outside its optimal range. Hybrid construction is therefore increasingly understood as a system level optimization strategy, rather than a transitional phase toward all-timber construction.

1.2. Advantages and Challenges of Wood-Based Hybrid Systems

From a structural perspective, numerous studies have demonstrated that wood-based hybrid systems can outperform mono-material timber structures in key performance criteria [19]. Timber–concrete composite floors have been reported to achieve approximately 30–60% higher bending stiffness in selected configurations, depending on connector type, span, and degree of composite action [20,21], while simultaneously improving acoustic insulation and vibration behavior. Similarly, timber–steel hybrid beams and frames have been shown to allow significant reductions in member dimensions compared to timber-only solutions under comparable load conditions, depending on reinforcement configuration and degree of composite action [22].

Environmental performance constitutes a further advantage. Life-cycle assessment studies have reported embodied-carbon reductions in the range of 20–50% for selected hybrid configurations, depending on system configuration, material quantities, and assessment boundaries [23]. On a material level, wood typically stores approximately 1t of CO_2 per cubic meter, while requiring significantly less primary energy for production compared to steel and concrete. Hybridization may enable these benefits to be realized while addressing project-specific requirements for structural robustness, fire safety, and regulatory approval.

From a construction and process perspective, wood-based hybrid systems support prefabrication and industrialized construction [24]. Several studies have reported reductions in on-site construction time under project-specific prefabrication conditions, while hybrid assemblies remain compatible with established concrete and steel construction practices [25]. This compatibility has been a key factor in the increasing adoption of hybrid systems in urban multi-story housing and mixed-use developments over the past decade.

Wood–hybrid construction, while advantageous in terms of carbon reduction and seismic performance, involves notable constraints. Structurally, EWPs can exhibit brittle failure modes, often requiring greater member depths and shorter spans than steel or reinforced concrete [26]. Achieving effective composite action between timber and concrete or steel can demand dense fastener layouts, increasing fabrication complexity and cost. Timber's susceptibility to moisture-related degradation, particularly in façade applications, necessitates careful detailing and planned maintenance [27]. Fire safety requirements frequently impose additional protective layers or charring allowances, adding cost and design constraints [28,29]. Economic viability may be affected by specialized connection systems and market perceptions regarding feasibility. Regulatory barriers, especially strict fire codes and limited guidance for novel systems, can further restrict application [30]. Moreover, environmental performance depends on material quantities and system configuration, requiring careful lifecycle assessment to avoid unintended impact trade-offs [31]. These disadvantages must be carefully managed to ensure the successful implementation of wood-hybrid construction projects.

The numerical ranges reported in this section should be interpreted as indicative values from individual studies rather than directly comparable pooled estimates. Differences in span, connector type, degree of composite action, load level, structural configuration, fire-protection strategy, prefabrication scope, and life-cycle assessment boundaries limit direct comparison between studies. Accordingly, these values are used in this paper to illustrate reported performance tendencies, not to establish universal quantitative benchmarks for wood-based hybrid construction.

1.3. Hybrid Construction within a Circular Economy Framework

The integration of hybrid construction with circular economy principles has become increasingly relevant as regulatory frameworks evolve. Circular construction emphasizes material efficiency, adaptability, design for disassembly, and end-of-life recovery. Wood-based hybrid systems are particularly compatible with these principles due to the demountability of timber components and the potential for mechanical connections [32].

However, recent research indicates that circular performance is highly sensitive to system configuration and detailing [33]. While timber elements can often be reused or recycled, concrete and steel components are typically downcycled or reprocessed with significant energy input. Hybrid construction mitigates this limitation by minimizing the volume of mineral-based materials and concentrating them in locations where reuse is less critical but performance demands are highest [34]. In this sense, hybrid construction represents a circular optimization strategy rather than a purely material-driven solution.

1.4. Research Gap and Aim of the Study

Despite the growing body of literature on wood-based composites and hybrid structural systems, much of the existing research remains focused on isolated components, laboratory-scale experiments, or single performance criteria, e.g., [35,36]. Fewer studies address hybrid construction as an integrated architectural and structural strategy aligned with circular economy objectives and contemporary regulatory requirements, particularly at the level of design-scale decision-making.

This study responds to this gap by presenting a structured practice- and education-oriented decision-support framework that synthesizes established wood-based hybrid construction solutions through a circular economy perspective. The aim is not to propose experimental materials, speculative systems, or a validated predictive model, but to document, systematize, and translate documented hybrid construction approaches into a practice-oriented reference that supports low-carbon building design, lifecycle-oriented decision-making, regulatory alignment, and architectural education.

The contribution of this study is therefore not the introduction of a new structural typology or a validated predictive model, but the development of an integrative three-level decision-support framework that organizes wood-based hybrid construction according to single component, system, and building scales. By linking each scale to structural performance, circular economy mechanisms, and early-stage design considerations, the framework clarifies how hybridization decisions can be understood and communicated across architectural and engineering practice.

2. Materials and Methods

2.1. Research Design and Methodological Approach

This research adopts a qualitative, non-experimental research design based on structured qualitative literature synthesis and applied framework development (Figure 1). The methodological approach is aligned with the nature of the research problem, which concerns the evaluation and organization of established wood-based hybrid construction solutions. The study does not aim to test new materials, validate prototypes, or generate primary experimental data. Instead, it consolidates and structures existing knowledge in a form that is analytically clear and directly applicable to architectural and construction practice.



Figure 1. Research process overview.

The choice of a non-experimental methodology reflects the maturity of the research field addressed. Wood-based hybrid construction has been extensively investigated through experimental, numerical, and applied studies over the past decades. As a result, the primary challenge at present is not the lack of performance data, but rather the fragmentation of knowledge across disciplines, scales, and application contexts. This study therefore positions itself at the level of systematization and synthesis, addressing how hybrid construction strategies can be understood, compared, and deployed within contemporary low-carbon and circular construction frameworks.

Accordingly, the research does not rely on primary experimental testing, full-scale measurements, or numerical simulations. Instead, it integrates secondary data derived from peer-reviewed journal articles, authoritative technical reports, and documented application examples. This methodological choice is consistent with established practices in architectural and construction research where system level performance, design logic, and regulatory compatibility are central concerns. By focusing on verified and documented sources, the study ensures that its findings are grounded in scientifically validated knowledge rather than speculative assumptions.

The overall research process was designed to improve transparency, traceability, and applicability. All methodological steps, from data selection to synthesis and framework development, were explicitly structured to allow the research logic to be followed and critically assessed. In addition, particular attention was given to ensuring that the outcomes of the methodology could be meaningfully applied both in professional practice and in architectural and construction education, reflecting the dual research and educational orientation of the underlying project.

2.2. Data Sources and Selection Criteria

The empirical basis of the study consists of a curated body of secondary data drawn from multiple categories of scientific and technical literature. The primary data sources include peer-reviewed journal articles focusing on wood-based hybrid construction, EWPs, and hybrid structural systems. These sources provide detailed discussions of material behavior, structural performance, and system integration strategies that form the technical backbone of the analysis.

In addition, state-of-the-art review papers addressing timber–concrete, timber–steel, and wood-based composite systems were included to ensure comprehensive coverage of established hybrid construction typologies. Such reviews are particularly valuable in identifying recurring performance trends, methodological approaches, and unresolved challenges across multiple studies. Authoritative technical publications and handbooks on wood-based materials, structural composites, and hybrid construction technology were also consulted, provided that these documents demonstrated clear methodological grounding and scientific credibility.

Documented application examples from academic and professional literature constituted a further data source. These examples were used not as isolated case studies, but as contextual references that illustrate how hybrid construction strategies are implemented in practice. Their inclusion supports the applied orientation of the research and enables cross-referencing between theoretical findings and real-world configurations.

The final synthesis retained 45 sources cited in the manuscript. These sources were identified through targeted keyword searches and citation tracking around wood-based hybrid construction, timber–concrete systems, timber–steel systems, wood-based composites, engineered wood products, circular economy, design for disassembly, embodied carbon, lifecycle assessment, prefabrication, digital construction, material passports, and multi-story timber buildings. The evidence base was further complemented with recent studies addressing sustainable construction, construction management, urban sustainability, material performance, and contemporary built-environment practice. The source-selection process was not intended to constitute a systematic review or meta-analysis; rather, it was designed as an applied qualitative synthesis using a purposive evidence base relevant to the three-level framework developed in this study.

To ensure methodological rigor, explicit selection criteria were applied to all potential sources. Only publications demonstrating scientific validity were included, meaning that they were indexed in recognized academic databases or issued by authoritative research institutions. Relevance was assessed based on the direct focus of the publication on wood-based hybrid systems at single component, system, or building scale. In addition, a clear performance orientation was required, with explicit discussion of structural, environmental, circular economy, or functional performance parameters.

Applicability to contemporary building practice constituted a further key criterion. Sources addressing outdated construction practices or contexts no longer relevant to current regulatory and urban conditions were excluded. Similarly, publications limited to speculative concepts, unverified prototypes, or purely aesthetic interpretations of hybridization were deliberately omitted to maintain analytical robustness and avoid methodological ambiguity. To reduce selection bias, evidence was drawn from multiple source types, including experimental studies, review articles, lifecycle-oriented studies, technical publications, and documented application examples. Findings were interpreted comparatively across sources rather than relying on single studies, and emphasis was placed on recurring patterns that could be linked directly to the three analytical levels of the proposed framework.

Because the reviewed literature differs in scope, methods, boundary conditions, and performance indicators, no numerical quality score or pooled statistical comparison was applied. Instead, the quality assessment focused on scientific credibility, methodological transparency, topical relevance, and applicability to contemporary design and construction practice. This approach is consistent with the aim of developing a practice-oriented decision-support framework rather than producing an exhaustive systematic review.

2.3. Analytical Framework: Hybridization Levels

To enable systematic comparison and synthesis across heterogeneous sources, wood-based hybrid construction systems were classified according to three hybridization levels, reflecting both the scale and complexity of material integration within the building (Figure 2). This classification constitutes the core analytical framework of the study and provides a coherent structure for organizing literature-derived findings alongside documented application examples. By distinguishing hybridization according to its spatial and functional scope, the framework supports consistent evaluation across diverse construction strategies [37].

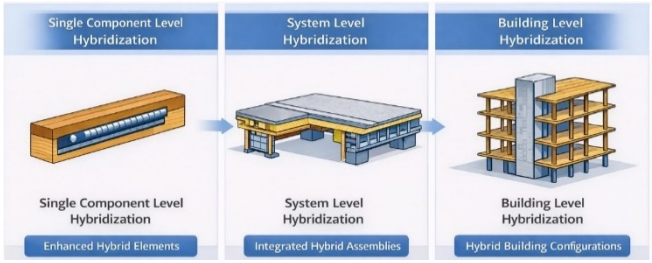


Figure 2. Hybridization level overview.

Single component level hybridization involves combining two or more materials within a single structural element to capitalize on their complementary mechanical properties. Typical examples include hybrid beams, columns, and wall panels with reinforcement embedded in an otherwise timber-based component. In timber–steel hybrids, steel enhances tensile capacity while timber provides compressive strength and constructability; in timber–concrete configurations, timber may contribute tensile performance and concrete supplies compressive resistance and stiffness. At this level, hybridization targets localized improvements in bending capacity, stiffness, durability, fire resistance, and moisture performance, while leaving the global structural system fundamentally unchanged.

System level hybridization involves integrating multiple structural components made of different materials to improve overall structural efficiency and performance. This approach applies to various building subsystems, including floors, frames, facades, cores, and primary load-bearing or lateral-resisting systems. Common examples include timber–concrete composite floors, where performance results from the interaction of multiple elements. At this level, connection detailing, load transfer mechanisms, and compatibility of material behavior under service and extreme conditions are critical. System level hybridization is widely applied in contemporary practice, particularly where timber superstructures are combined with incombustible core systems, such as concrete, to satisfy fire regulations while remaining compatible with established construction processes.

Building level hybridization involves the strategic combination and spatial distribution of different structural systems within a building to enhance adaptability, structural performance, and regulatory compliance. Common applications include reinforced concrete cores combined with timber superstructures, as well as timber-based extensions added to existing or historical buildings, where wood is often preferred due to its low self-weight. At this level, hybridization addresses global structural behavior, fire safety, lateral stability, stiffness requirements, and code compliance, particularly in multi-story and tall buildings. The analytical focus lies in understanding how building level hybrid configurations enable the use of timber in building typologies where purely timber-based solutions would otherwise be constrained.

The adoption of this three-level framework avoids oversimplified material comparisons and reflects the actual decision-making scales encountered in architectural and structural design practice. Moreover, it enables the identification of systematic relationships between hybridization level, performance outcomes, and circular economy potential. In particular, the framework highlights how increasing the scale of hybridization shifts the emphasis from localized material enhancement toward system integration, regulatory alignment, and lifecycle considerations.

2.4. Evaluation Parameters

Across all hybridization levels, the analysis employed a consistent set of performance-oriented evaluation parameters derived from recurring criteria in scientific literature. Structural performance was assessed in terms of load-bearing capacity, stiffness, span capability, and serviceability behavior, with particular attention to how hybridization enables performance improvements or material efficiency gains.

Fire safety considerations formed a further evaluation dimension, focusing on material behavior under fire exposure, protection strategies, and implications for regulatory compliance. The analysis examined how hybrid configurations mitigate limitations associated with pure timber systems, especially in relation to fire resistance requirements in multi-story construction.

Environmental performance was evaluated with emphasis on embodied carbon reduction potential, material efficiency, and the carbon storage capacity of wood-based components. Hybridization was considered in terms of its ability to strategically reduce the use of emission-intensive materials while maintaining structural reliability and durability.

Construction and process-related aspects were also examined, including prefabrication potential, construction speed, and compatibility with established construction practices. These factors are critical in assessing the feasibility and scalability of hybrid systems in real-world projects.

Finally, circular economy potential was assessed through attributes such as adaptability, design for disassembly, reuse potential, and end-of-life scenarios. Attention was given to the role of system configuration and connection strategies in enabling or constraining circular performance. Rather than assigning numerical scores or weights, the evaluation focused on comparative performance tendencies and qualitative relationships, reflecting the diversity of systems, contexts, and boundary conditions reported in the literature.

2.5. Synthesis and Framework Development

The final stage of the methodology involved qualitative synthesis, through which recurring patterns, advantages, and limitations across hybridization levels were identified. This synthesis formed the basis for the development of a structured reference framework linking wood-based hybrid construction strategies with circular economy principles.

The framework was refined through iterative comparison between literature-derived findings and documented application examples. Emphasis was placed on identifying robust and transferable design logics rather than context-specific optimizations. The resulting framework is intended to function as a decision-support reference for architects and engineers, a teaching and learning resource for architectural and construction education, and a conceptual bridge between research findings and regulatory or professional practice.

2.6. Circular Economy Logic Embedded in the Hybridization Framework

The circular economy logic adopted in this study builds on established research that conceptualizes circularity in the built environment as a function of system organization, adaptability, and value retention rather than material selection alone [38]. Previous studies emphasize that circular performance in buildings is primarily determined by early-stage design decisions related to modularity, reversibility, and lifecycle planning, rather than by the isolated environmental

properties of individual materials [39]. In line with this literature, the proposed hybridization framework embeds circular economy principles directly within structural decision-making across multiple design scales.

At the single component level, circular economy potential is predominantly associated with durability-oriented strategies and material efficiency. Research on timber–concrete and timber–steel hybrid systems demonstrates that localized reinforcement and composite action can enhance structural capacity and serviceability, thereby reducing material demand in selected applications [20–22,26]. However, circular economy literature on the built environment indicates that component-level hybridization often limits end-of-life separation and high-value recovery due to irreversible material integration [37,40]. As a result, circularity at the single component level is primarily achieved through life-extension rather than reuse or material recirculation.

At the system level, circular economy performance becomes more explicit through modular assemblies, standardized interfaces, and partial disassembly. Studies framing buildings as material banks show that system level configurations enable selective replacement, upgrading, and reuse of subsystems while preserving overall structural integrity. In this context, circular outcomes depend strongly on connection logic and load-transfer mechanisms, which determine whether hybrid assemblies can be separated into recoverable components at later stages of the building lifecycle [37].

At the building level, circularity is most strongly influenced by spatial organization, system zoning, and long-term adaptability. Research on building lifecycle behavior highlights that structures designed to accommodate change over time retain functional and material value across multiple life cycles, particularly when primary structural systems are separated from adaptable secondary systems. Hybrid configurations that combine timber superstructures with concrete, steel, or other complementary primary systems may therefore support circular economy objectives by enabling functional transformation, regulatory alignment, and future reuse scenarios when reversibility, adaptability, and lifecycle documentation are considered from the early design stage.

By grounding circular economy logic in these established research findings, the proposed framework clarifies that higher levels of hybridization do not inherently result in improved circular performance. Instead, circular outcomes depend on the alignment between hybridization scale, system configuration, and anticipated lifecycle pathways, consistent with contemporary circular economy frameworks for the built environment [38]. Accordingly, building level hybridization should be understood as potentially offering broader strategic leverage for circular economy implementation when supported by system separation, reversibility, adaptability, and lifecycle documentation, rather than as automatically producing superior circular outcomes.

On this basis, Section 3 applies the circular economy logic and the three-level hybridization framework as the organizing structure for the reported findings. The results are first presented through recurring hybridization patterns identified in the reviewed literature and documented application examples. They are then organized according to the single component level, system level, and building level to show how each level relates to structural performance, system integration, circular economy mechanisms, and early-stage design decision-making. The section concludes with a cross-level synthesis and practical interpretation of the framework. This structure clarifies how the analytical framework developed in the Method section is operationalized in the Results section, while also making clear that the findings represent a qualitative synthesis rather than statistical comparison or experimental validation.

3. Results

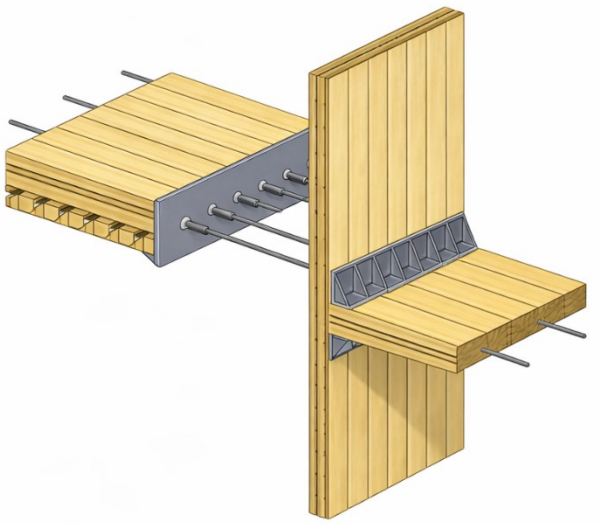
3.1. Overview of Identified Hybridization Patterns

Application of the proposed analytical framework to the reviewed literature and documented application examples revealed consistent and recurring hybridization patterns across all three levels. Rather than presenting isolated technical solutions, the results demonstrate that wood-based hybrid construction is predominantly employed as a performance-driven and context-sensitive strategy. Across the dataset, hybridization was not used to maximize the absolute share of timber, but to allocate materials rationally according to structural, regulatory, and lifecycle requirements.

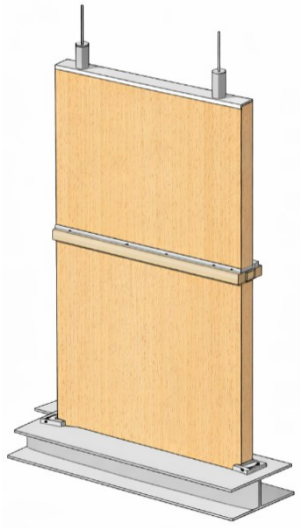
A key overarching result is that the scale of hybridization directly influences both the type of performance gains achieved and the complexity of system integration. Lower hybridization levels primarily address localized mechanical limitations of timber, whereas higher levels increasingly respond to global structural behavior, fire safety regulations, and long-term adaptability. These relationships are articulated in detail in the following subsections.

3.2. Results at the Single Component Level

At the single component level, hybridization is consistently applied to enhance the performance of individual timber elements through targeted modifications, without altering the overall structural system. The reviewed studies demonstrated that reinforcing timber beams, columns, or slabs with steel bars, fiber-reinforced polymers, or composite layers leads to measurable improvements in bending capacity, stiffness, and resistance to cracking and long-term deformation (Figure 3). Single component level hybridization also includes several special applications, e.g., wood-plastic composites (Figure 4).



(a)



(b)

Figure 3. (a) Post-tensioned timber slab (Adapted from [41]); (b) Post-tensioned timber shear wall (Adapted from [42]).

Figure 3 illustrates the single component level logic of hybridization, where performance is improved within the boundaries of an individual timber element rather than through changes to the whole structural system. The examples demonstrate how localized reinforcement can enhance stiffness, strength, or lateral resistance, but they also highlight a circular economy trade-off: when different materials are integrated within a single member, later separation, reuse, and high-value recovery may become more difficult.



Figure 4. Wood-plastic composite applications. (Adapted from [43])

Figure 4 extends the single component level discussion beyond load-bearing reinforcement by showing how wood-based composites may improve durability and material efficiency in specific applications. However, such examples also demonstrate that material hybridization does not automatically improve circularity, because the integration of polymers and wood fibers can complicate separation and high-value recycling at the end of life.

Across the examined sources, single component level hybridization was most frequently employed to address serviceability-related limitations, such as excessive deflection or vibration sensitivity. By selectively reinforcing critical zones of timber elements, designers were able to reduce cross-sectional dimensions while maintaining or improving structural performance. Several studies also reported improved durability and fire performance when hybrid reinforcement strategies were combined with appropriate detailing and protective measures.

From a system integration perspective, single component level hybridization was found to be minimally disruptive. Because material integration occurs within individual elements, this approach does not require changes to structural layout, construction sequencing, or regulatory classification. As a result, it is particularly suitable for retrofit projects and for applications where design flexibility is constrained.

However, the results also indicate that the circular economy potential at this level is limited. The intimate integration of dissimilar materials within a single component complicates disassembly and material separation at the end of life. While durability improvements may extend service life, opportunities for reuse or high-quality recycling remain restricted unless reversible reinforcement strategies are employed.

3.3. Results at the System Level

The system level emerged as the most prevalent and technologically mature form of wood-based hybrid construction in the reviewed literature. At this level, hybridization occurs through the interaction of multiple structural components that together form a functional subsystem. Common examples include timber–concrete composite floors (Figure 5), timber–steel composite beam assemblies (Figure 6), and timber–concrete composite wall applications (Figure 7).

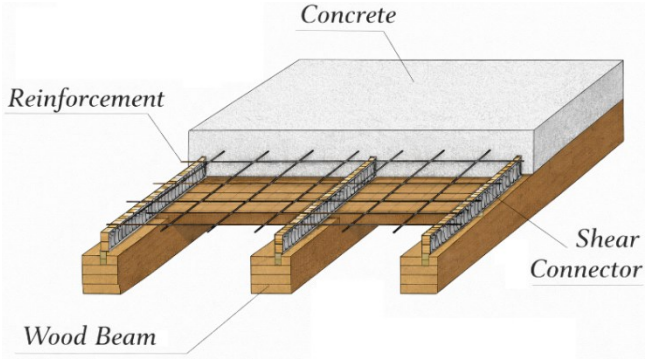


Figure 5. Timber–concrete composite floor. (Adapted from [20])

Figure 5 represents system level hybridization because performance depends on the interaction between timber, concrete, and the shear connection rather than on either material alone. Within the proposed framework, this example illustrates why connection design is central to both structural performance and circular economy potential: the same interface that enables composite action also determines whether the assembly can later be separated, repaired, or partially reused.



Figure 6. Timber–steel composite beam.

Figure 6 further illustrates the system level logic of hybridization by showing how timber and steel can be combined to balance low self-weight, tensile capacity, ductility, and constructability. Its relevance to the framework lies in demonstrating that system level performance is not produced by material substitution alone, but by the coordinated behavior of components and connections.



Figure 7. Timber–concrete composite wall.

Figure 7 shows that system level hybridization can also occur in wall assemblies, where structural, fire-safety, durability, and envelope-related functions may overlap. This supports the framework's emphasis on subsystem coordination and demonstrates that circular economy outcomes depend on whether layers, interfaces, and connections are designed for maintenance, replacement, or disassembly.

The results indicate that performance gains at this level arise primarily from composite action and coordinated system behavior rather than from isolated component enhancements. Timber–concrete composite floors (Figure 5) consistently demonstrated increased stiffness, improved vibration performance, and enhanced acoustic behavior compared to pure timber floor systems. Similarly, timber–steel frame systems benefited from improved tensile capacity and ductility while retaining the reduced self-weight characteristic of timber structures.

Connection detailing was identified as a decisive factor influencing system performance at the system level. The reviewed studies repeatedly emphasized that load transfer mechanisms, slip behavior, and differential material response under moisture and temperature variations significantly affect both short- and long-term performance. Systems employing well-defined mechanical connections and controlled composite action were found to exhibit reliable behavior under service and extreme loading conditions.

In terms of construction and process-related outcomes, the system level hybridization was associated with high compatibility with prefabrication and industrialized construction methods. Many documented systems enabled off-site fabrication of timber components combined with on-site casting or assembly of complementary materials. This approach supports reduced construction time and improved quality control while remaining compatible with established construction practices.

From a circular economy perspective, the results indicate a moderate to high potential at this level. Systems designed with mechanical connections and modular assemblies allow partial disassembly, component replacement, and adaptability. However, the extent of circularity remains strongly dependent on early design decisions, particularly regarding connection strategies and construction sequencing.

3.4. Results at the Building Level

At the building level, hybridization was primarily driven by global performance requirements and regulatory constraints rather than localized material optimization. The reviewed examples commonly employed reinforced concrete cores combined with timber superstructures, podium–tower configurations, or material zoning based on height and functional requirements.

The results demonstrate that building level hybridization can play a critical role in enabling timber use in multi-story and tall buildings, where pure timber systems may face limitations related to fire resistance, lateral stability, or stiffness. Ascent in Milwaukee provides a representative example of this design logic, as the building illustrates how timber can be combined with complementary structural systems to address building scale performance and regulatory requirements (Figure 8). By concentrating concrete or steel elements in zones requiring high mass, rigidity, or fire resistance, these configurations allow extensive use of timber in gravity load-bearing systems and building envelopes.



Figure 8. Ascent in Milwaukee as a representative building level timber-hybrid example, illustrating how material allocation at the building scale is shaped by global structural behavior, fire-safety requirements, constructability, and regulatory constraints. Courtesy of Thornton Tomasetti.

Figure 8 illustrates the building level dimension of the framework, where the design question shifts from improving individual members or subsystems to allocating materials across the whole building. At this scale, hybridization becomes relevant because fire safety, lateral stiffness, constructability, regulatory alignment, and long-term adaptability must be coordinated through building scale material zoning.

Performance outcomes at this level were predominantly evaluated in terms of global structural behavior, regulatory compliance, and constructability. The reviewed literature indicates that hybrid building systems may support regulatory approval processes by combining timber use with complementary materials that address fire safety, stiffness, robustness, and constructability requirements. However, embodied-carbon benefits remain project-specific and depend on material quantities, system configuration, and life-cycle assessment boundaries.

From a lifecycle perspective, building level hybridization may offer broad strategic opportunities for circular economy implementation, but only when early-stage design coordination addresses system separation, adaptability, maintenance, and future reuse scenarios. Clear separation between primary load-bearing systems and replaceable secondary components was identified as a key factor enabling adaptability, functional change, and future reuse.

3.5. Cross-Level Comparison and Synthesis of Findings

Comparative analysis across the three hybridization levels reveals a progression in both design complexity and potential impact. At the single component level, hybridization provides targeted performance enhancement with limited disruption to the overall structural system. However, its circular economy

potential remains constrained when different materials are irreversibly integrated within the same element.

At the system level, hybridization offers broader performance benefits through composite action, modular coordination, and improved interaction between structural components. This level provides stronger opportunities for partial disassembly, replacement, and upgrading, provided that connection strategies are designed with reversibility and lifecycle planning in mind.

At the building level, hybridization addresses global structural behavior, fire safety, regulatory alignment, and long-term adaptability. Its potential impact is therefore broader, but it also requires greater coordination between architectural, structural, regulatory, and construction-related decisions.

Across all levels, the findings indicate a shift from simple material substitution toward strategic material allocation. Successful wood-based hybrid solutions do not aim to maximize timber use indiscriminately. Instead, they position timber, concrete, steel, and other materials according to their structural, environmental, regulatory, and lifecycle functions.

To clarify the practical use of the proposed framework, Table 1 illustrates how the three levels can support early-stage design decision-making. The table is not intended as a validated case study, but as an application-oriented synthesis linking design use, circular economy implications, and key cautions.

Table 1. Illustrative use of the three-level framework as an early-stage design decision tool

Framework level	Early-stage design use	Circular economy implication and caution
Single component level	Strengthening or protecting an individual timber member, such as a reinforced beam, slab, column, or wall panel	Mainly supports service-life extension and material efficiency. End-of-life separation may be difficult if materials are irreversibly integrated.
System level	Coordinating multiple components as a functional subsystem, such as a timber–concrete floor, timber–steel beam, or hybrid wall assembly	Can support partial disassembly, component replacement, and modular upgrading. Circularity depends on reversible connections, sequencing, and documentation.
Building level	Allocating materials across the whole building, such as timber superstructures combined with concrete or steel cores, podiums, or zoning strategies	Can support adaptability, system separation, future transformation, and reuse. Benefits depend on early coordination, regulatory verification, and lifecycle planning.

This illustrative application shows that the framework is most useful when it structures design reasoning rather than prescribing a single optimal hybrid solution.

3.6. Implications for Circular Economy-Oriented Hybrid Construction

The results suggest that circular economy performance in wood-based hybrid construction depends primarily on system configuration, connection logic, and lifecycle planning rather than material choice alone. At the single component level, circularity is mainly linked to durability and service-life extension, although intimate material integration can limit later separation and reuse. At the system level, modular assemblies and mechanical connections can support partial disassembly, replacement, and adaptation when reversibility and documentation are considered early. At the building level, circular economy opportunities depend on strategic material zoning, separation between primary and secondary systems, and anticipated future transformation scenarios.

Overall, the findings indicate that circularity should be treated as a design-dependent outcome. Wood-based hybrid construction can support circular economy objectives when reversibility, modularity, system separation, documentation, and end-of-life pathways are integrated into the design process.

4. Discussion

4.1. Interpreting Hybridization as a Performance-Oriented Design Strategy

The results indicate that wood-based hybrid construction should be understood as a scale-dependent design strategy rather than as simple material substitution. Timber, concrete, steel, and other materials are allocated according to their structural, regulatory, environmental, and lifecycle functions.

At lower hybridization levels, performance gains are achieved mainly through localized improvement of timber elements. At system and building levels, hybridization operates at broader decision scales, where fire safety, global structural behavior, constructability, and regulatory alignment can be coordinated more holistically. This does not mean that higher levels are automatically superior; rather, each level addresses different design problems.

4.2. Hybridization Scale and Its Implications for Circular Economy Objectives

The findings suggest a conditional relationship between hybridization scale and circular economy potential. At the single component level, durability gains may extend service life, but they should not automatically be equated with circularity because reuse and separation can remain limited. At the system level, circular economy opportunities depend largely on reversible connections, modularity, construction sequencing, and documentation.

At the building level, hybridization can support circular economy objectives through material zoning and hierarchical system separation. However, higher hybridization levels should not be understood as inherently more circular, particularly because the present study does not include comparative lifecycle assessment. Circular economy value depends on whether adaptability, reversibility, system separation, and lifecycle documentation are deliberately integrated into the design process.

4.3. Regulatory Context and Practical Relevance

In regulatory contexts where building level carbon reporting and embodied-carbon requirements are becoming more important, hybrid construction may provide a pragmatic pathway for aligning timber use with emerging carbon-related requirements without requiring radical departures from established construction practices [44,45]. Building level hybridization may allow timber to be used more extensively while supporting project-specific verification of fire safety, stiffness, robustness, and code-related requirements.

The results suggest that hybrid construction may function as an enabling strategy for broader adoption of wood-based materials in contexts where pure timber solutions remain constrained. Rather than framing hybrid systems as a second-best alternative, the findings suggest that they can be interpreted as a mature and scalable approach when regulatory requirements, environmental objectives, and lifecycle considerations are addressed together.

4.4. Implications for Architectural and Structural Design Practice

From a design perspective, the study highlights the importance of aligning hybridization strategies with clearly defined performance objectives. Architects and engineers are encouraged to move beyond material-centric narratives and instead adopt a systemic design logic that integrates structural performance, constructability, regulatory compliance, and lifecycle considerations.

The analytical framework developed in this study provides a structured lens through which hybridization decisions can be evaluated at different scales. By explicitly distinguishing between Single component, system, and building level hybridization, the framework supports informed decision-making and facilitates communication between architectural and engineering disciplines. This clarity is particularly relevant in collaborative design processes where hybrid solutions are negotiated across multiple stakeholders.

4.5. Limitations and Scope of Interpretation

While the findings provide valuable insights into wood-based hybrid construction, they should be interpreted within the methodological scope of the study. The research is based on secondary data synthesis and documented application examples rather than primary experimental testing. As such, the results do not claim to quantify performance outcomes, establish causal relationships, or provide statistically generalizable conclusions. Rather, the study aims to identify recurring patterns and transferable design logics within the reviewed body of literature.

The proposed three-level framework should therefore be understood as an interpretive and decision-support structure rather than as a validated classification model. Its value lies in clarifying the scale at which hybridization decisions are made and in linking these scales to structural, regulatory, and circular economy considerations. However, the framework has not been validated through expert assessment, comparative testing against alternative classifications, or application to a full design case study.

In addition, the study does not constitute a systematic review or meta-analysis. The reviewed sources differ in scope, methods, boundary conditions, and performance indicators, which limits direct comparison between reported outcomes. For this reason, the numerical ranges and performance tendencies discussed in the manuscript should be interpreted as indicative findings from selected studies rather than universal benchmarks.

Future research could build upon this framework by integrating quantitative lifecycle assessment, experimental validation of specific hybrid configurations, expert-based evaluation of the proposed classification, comparative analysis with existing hybrid construction frameworks, or post-occupancy evaluation of hybrid buildings. Nevertheless, the present study demonstrates that meaningful contributions to low-carbon and circular construction discourse can be achieved through structured synthesis and applied framework development, provided that the findings are interpreted as a conceptual and practice-oriented reference rather than as a predictive or universally validated model.

5. Conclusion

This paper examined wood-based hybrid construction through a structured analytical framework grounded in circular economy principles. By synthesizing established scientific literature and documented application examples, the study suggests that hybrid construction should not be interpreted as a compromise solution, but as a scale-dependent design strategy for allocating materials according to structural, regulatory, environmental, and lifecycle requirements.

The proposed three-level framework, consisting of the single component level, system level, and building level, provides a coherent lens for evaluating hybrid construction strategies across different design scales. The findings suggest that each level addresses different design problems: localized member performance at the single component level, subsystem coordination at the system level, and strategic material allocation at the building level.

The framework's main value lies in organizing fragmented knowledge into a practice- and education-oriented decision-support structure. It can support communication between architectural and structural design disciplines and help integrate circular economy considerations into early-stage decision-making. However, the framework should be interpreted as a conceptual and practice-oriented reference rather than as a validated predictive model. Future research should further test its applicability through comparative lifecycle assessment, expert evaluation, and full design case studies.

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Author contributions

Markku Karjalainen: Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Writing-Reviewing and Editing, Supervision, Project administration, Funding acquisition. **Hüseyin Emre Ilgın:** Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Writing-Original draft preparation, Writing-Reviewing and Editing, Visualization. **Teemu Hirvilammi:** Conceptualization, Methodology, Writing-Reviewing and Editing, Supervision, Project administration. **Sami Pajunen:** Conceptualization, Methodology, Writing-Reviewing and Editing, Supervision, Project administration.

Conflicts of interest

The authors declare no conflicts of interest.

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